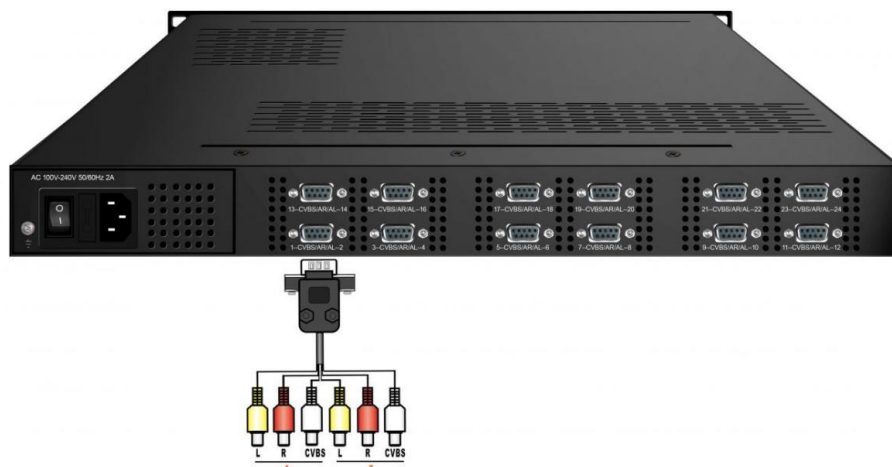


FMUSER FUQ-4653XY Encoder Modulator 24 CVBS & 128 IP Inputs DVB-C-T SD MPEG-2 H.264 Integrated Multiplexing Scrambling for CATV IPTV





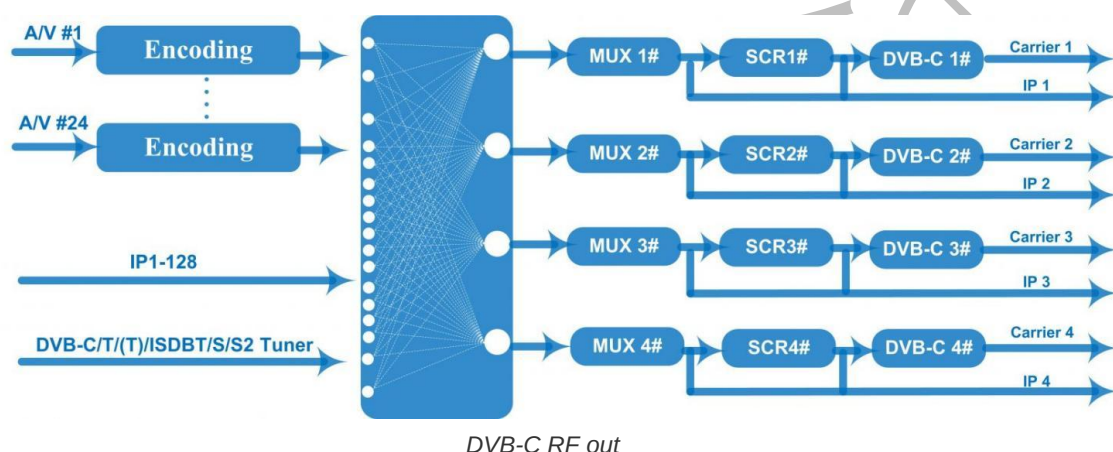
Product Overview

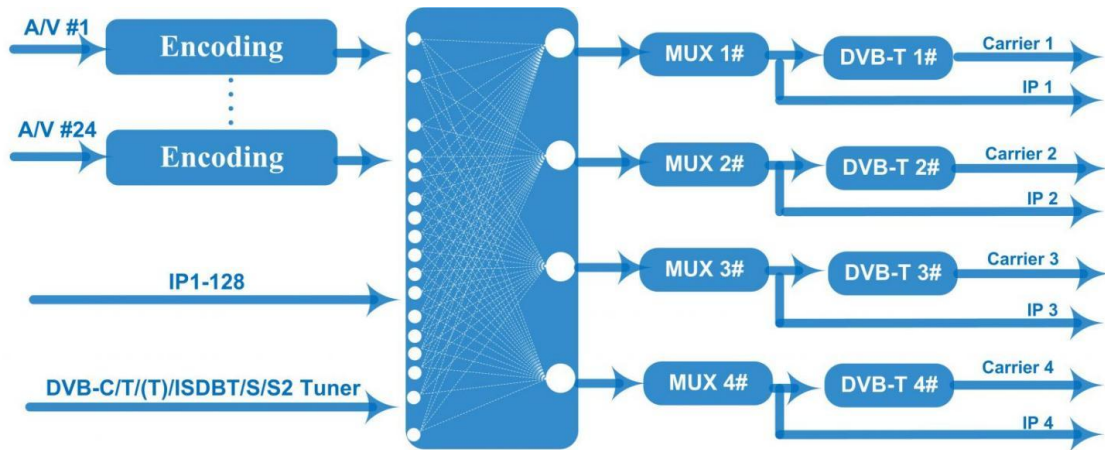
The FMUSER FUQ-4653LM integrates SD encoding, multiplexing, scrambling, and modulation, supporting up to 24 CVBS inputs, an optional DVB-C/T/ISDBT/S/S2 tuner, and 128 IP inputs via Data1 (GE) and Data2 (FE) ports. It processes signals through encoding and modulation, delivering 4 DVB-C/T RF outputs and 4 MPTS IP streams. Managed via a web interface, this device is ideal for CATV headends and hospitality TV systems.

Key feature

- Supports multi-format encoding with MPEG-2/H.264 video and MPEG1 Layer 2/LC-AAC/HE-AAC audio for up to 24 CVBS inputs.
- Integrates optional DVB-C/T/ISDBT/S/S2 tuner and 128 UDP/RTP IP inputs via dual Ethernet ports.
- Delivers 4 DVB-C/T RF outputs (50-960MHz) and 4 MPTS IP streams over UDP/RTP.
- Features advanced signal control including PID remapping, PCR adjustment, PSI/SI editing, and logo/QR code insertion.
- Enables web-managed operations for remote configuration, scrambling control (DVB-C only), and firmware updates.

Principle Chart





DVB-T RF out

Technical specification

Input	8/12/16/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT/S/S 2 (Optional) Tuner for re-mux, F type interface 128 IP input over UDP and RTP through DATA1 and DATA2, RJ45	8/12/16/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT/S/S 2 (Optional) Tuner for re-mux, F type interface 128 IP input over UDP and RTP through DATA1 and DATA2, RJ45	8/12/16/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT/S/S 2 (Optional) Tuner for re-mux, F type interface 128 IP input over UDP and RTP through DATA1 and DATA2, RJ45	8/12/16/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT/S/S 2 (Optional) Tuner for re-mux, F type interface 128 IP input over UDP and RTP through DATA1 and DATA2, RJ45	8/12/16/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT/S/S 2 (Optional) Tuner for re-mux, F type interface 128 IP input over UDP and RTP through DATA1 and DATA2, RJ45
Video	Resolution	720*576@50i	720*576@50i	720*576/352* 288/320*240/ 320*180/ 176 *144/160*120 /160*90@50H Z	720*576/352* 288/320*240/ 320*180/ 176 *144/160*120 /160*90@50H Z
Video	Resolution	720*480@60i	720*480@60i	720*480/352* 288/320*240/ 320*180/ 176 *144/160*120 /160*90@60H Z	720*480/352* 288/320*240/ 320*180/ 176 *144/160*120 /160*90@60H Z

Video	Encoding	MPEG-2, MPEG4 AVC/H.264	MPEG-2, MPEG4 AVC/H.264	MPEG-2, MPEG4 AVC/H.264	MPEG-2, MPEG4 AVC/H.264
Video	Bit-rate	0.5~5Mbps	0.5~5Mbps	0.5~5Mbps	0.5~5Mbps
Video	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video	GOP Structure	IPPP, IBPBP, IBBPB, IBBBP	IPPP, IBPBP, IBBPB, IBBBP	IPPP, IBPBP, IBBPB, IBBBP	IPPP, IBPBP, IBBPB, IBBBP
Video	Advanced Pretreatment	De-interlacing , noise reduction	De-interlacing , noise reduction	De-interlacing , noise reduction	De-interlacing , noise reduction
Audio	Encoding	MPEG1 Layer 2, LC-AAC, HE-AAC	MPEG1 Layer 2, LC-AAC, HE-AAC	MPEG1 Layer 2, LC-AAC, HE-AAC	MPEG1 Layer 2, LC-AAC, HE-AAC
Audio	Sampling rate	48KHz	48KHz	48KHz	48KHz
Audio	Bits per sample	32-bit	32-bit	32-bit	32-bit
Audio	Bit-rate	48Kbps~384K bps (each channel)	48Kbps~384K bps (each channel)	48Kbps~384K bps (each channel)	48Kbps~384K bps (each channel)
Multiplexing	Maximum PID Remapping	128 PID input per channel	128 PID input per channel	128 PID input per channel	128 PID input per channel
Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Scrambling (Only for DVB-C RF out)	Maximum simulcrypt CA	4	4	4	4
Scrambling (Only for DVB-C RF out)	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B

Scrambling (Only for DVB-C RF out)	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation	DVB-C	RF out	RF out	4*RF DVB-C out (4 carriers combined output)	4*RF DVB-C out (4 carriers combined output)
Modulation	DVB-C	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	DVB-C	MER	MER	≥40dB	≥40dB
Modulation	DVB-C	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-C	RF output level	RF output level	-25~-1dBm, 1dBm step	-25~-1dBm, 1dBm step
Modulation	DVB-C	Symbol Rate	Symbol Rate	5.0Msps~7.0 Msps, 1ksps step	5.0Msps~7.0 Msps, 1ksps step
Modulation	DVB-C	Constellation	Constellation	16/32/64/128/ 256QAM	16/32/64/128/ 256QAM
Modulation	DVB-C	Constellation	Constellation	J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	16/32/64/128/ 256QAM	64/256 QAM
Modulation	DVB-C	Bandwidth	Bandwidth	8M	6M
Modulation	DVB-T	RF out	RF out	4*RF DVB-T out (4 carriers combined output)	4*RF DVB-T out (4 carriers combined output)
Modulation	DVB-T	Standard	Standard	EN300744	EN300744
Modulation	DVB-T	MER	MER	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF output level	RF output level	-28~ -3 dBm, 0.1dBm step	-28~ -3 dBm, 0.1dBm step
Modulation	DVB-T	FFT mode	FFT mode	2K,	2K,

Modulation	DVB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulation	DVB-T	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Stream output	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)
Stream output	4 IP MPTS output over UDP/RTP through Data1 (1000M Base-T Ethernet interface)	4 IP MPTS output over UDP/RTP through Data1 (1000M Base-T Ethernet interface)	4 IP MPTS output over UDP/RTP through Data1 (1000M Base-T Ethernet interface)	4 IP MPTS output over UDP/RTP through Data1 (1000M Base-T Ethernet interface)	4 IP MPTS output over UDP/RTP through Data1 (1000M Base-T Ethernet interface)
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)
System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×410mm×44mm	482mm×410mm×44mm	482mm×410mm×44mm
Miscellaneous	Approx weight	Approx weight	8kg	8kg	8kg
Miscellaneous	Environment	Environment	0~45℃(work) -20~80℃(storage)	0~45℃(work) -20~80℃(storage)	0~45℃(work) -20~80℃(storage)
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz

Order Information

	FU-4653H B-8 (DVB-C RF out)	FU-4653B- 8 (DVB-T RF out)	FU-4653H B-12 (DVB-C RF out)	FU-4653H B-12 (DVB-T RF out)	FU-4653H B-24 (DVB-C RF out)	FU-4653H B-24 (DVB-T RF out)
8 CVBS Input	√	√				
12 CVBS Input			√	√		
24 CVBS Input					√	√
Scrambling function	√		√		√	

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)